GreenEye: Development of Real-Time Traffic Signal Recognition System for Visual Impairments

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Research Background

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For **people those that have disabled vision, it is a big challenge to walk across the crosswalks.**[3] It is challenging for them to check the color of the traffic light or check the remaining crossing time. Further, **only 34% of the total traffic lights in Korea has the beacons** **and** **many of them are broke**[1].

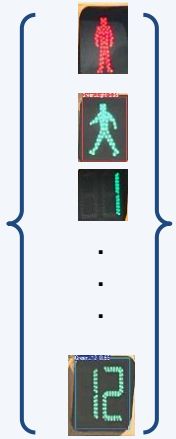
In 2019, Handong university in Korea conducted a research about recognizing the Green and Red traffic light. They researched to develop a machine learning algorithm that recognizes pedestrian signals in real-time.[4]

But the **previous research was about developing a software that only recognizes green or red color of the pedestrian signals. Further, it used YOLOv2, and older YOLO model with lower accuracy and training abilities. My research is about developing a software that recognizes only the green and red signals but also how many seconds the pedestrians are left with to cross the crosswalk, utilizing YOLOv5, an advanced YOLO model**.

[1] 기자 박성용. (2021, October 15). 신호등 설치 횡단보도 3곳 중 2곳, 시각장애인 음향신호기 없어. 웰페어뉴스. http://www.welfarenews.net/news/articleView.html?idxno=79056
[2] 기자 오경민. (2021, May 3). 시각장애인은 오늘도 목숨 걸고 길을 건넌다. 경향신문. https://m.khan.co.kr/national/national-general/article/202105032146015#c2b
[3] 시각장애인을 위한 초록이 들리는 신호등. 네이버/보건복지부 공식 블로그. (2017, October 27). https://m.blog.naver.com/PostView.naver?isHttpsRedirect=true&blogId=mohw2016&logNo=221126430901
[4] Oh, S., Jeong, K., Kim, H., & Kim, Y.-K. (2019). Development of street crossing assistive embedded system for the visually-impaired using machine learning algorithm. *Journal of the HCI Society of Korea*, 14(2), 41–47. https://doi.org/10.17210/jhsk.2019.05.14.2.41

Proposed GreenEye System

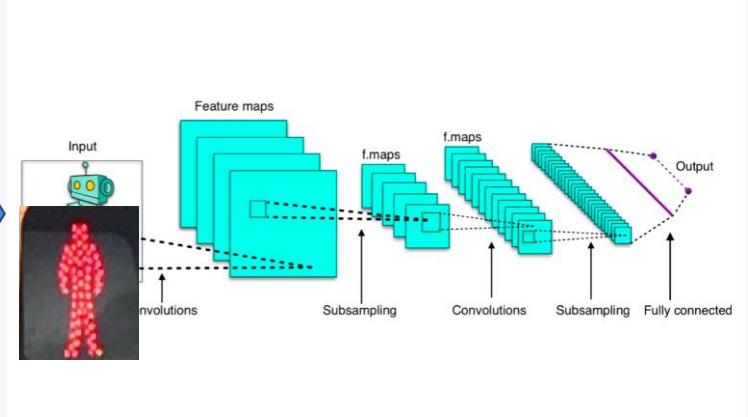
Traffic Light Photo Taking



Traffic Light & Number DB

(Train & Test Data)

Data Labeling



YOLOv5 Google Colaboratory CO

Yolo v5-Deep Learning Based Traffic Light & Number Recognition Training

Traffic Light & Number Recognition

GreenEye Mount

"Seven seconds left."

I want to **help the visually impaired people to cross the crosswalk.** The **proposed system roots on deep learning, while being able to recognize not only Red Light and Green Light** but also **how much time is left for the green light in real time**.

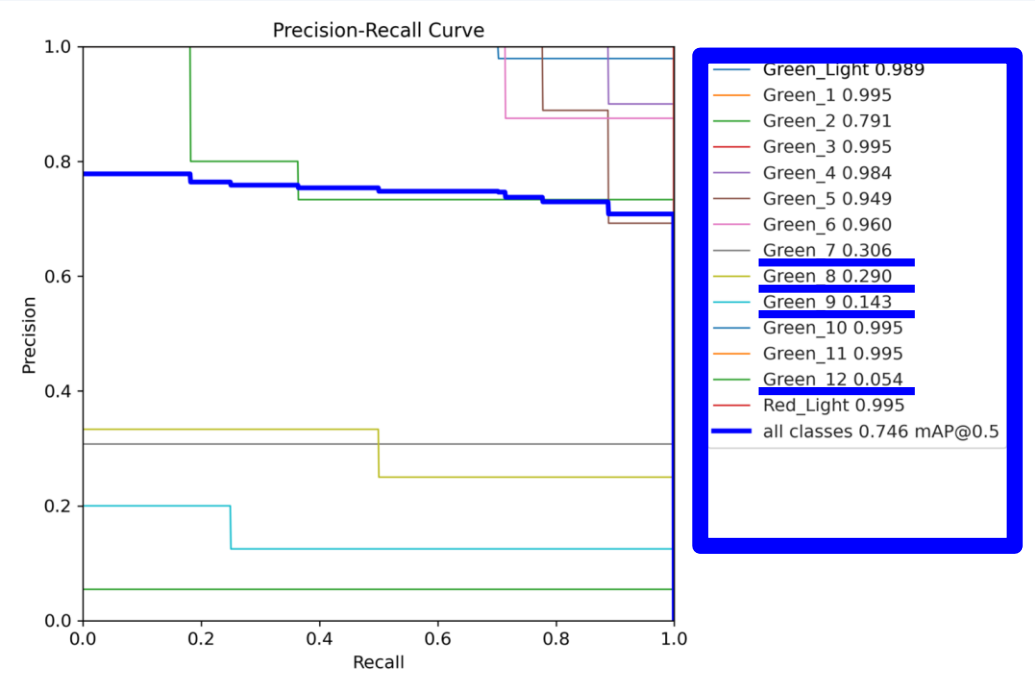
Experimental Results

- Experimental Setting & 9:1 Ratio Training

Classes	# of Data
Green_1	57
Green_2	77
Green_3	79
Green_4	82
Green_5	79
Green_6	72
Green_7	51
Green_8	41
Green_9	34
Green_10	44
Green_11	44
Green_12	24
Green Light	100
Red Light	100
Total	**884**

I used total of 14 classes: Green and Red light + Green_1 to Green_12. and total of 884 images to train and validate the YOLO traffic light recognition model.

The average precision for the whole classes was 0.746. This means the model was correct in recognizing the traffic light images 74.6% of the time. The model improved again. Green_7, Green_8, Green_9 and Green_12 again were the classes that had the lowest precision rate of less than 0.4.



Experimental Results

- Data Imbalance

Classes	# of Data
1	57
2	77
3	79
4	82
5	79
6	72
7	51
8	41
9	34
10	44
11	44
12	24
Green Light	100
Red Light	100
Total	884

Classes	# of Data
1	66
2	77
3	79
4	82
5	79
6	72
7	89
8	80
9	76
10	85
11	78
12	49
Green Light	100
Red Light	100
Total	1113

Data Imbalance:

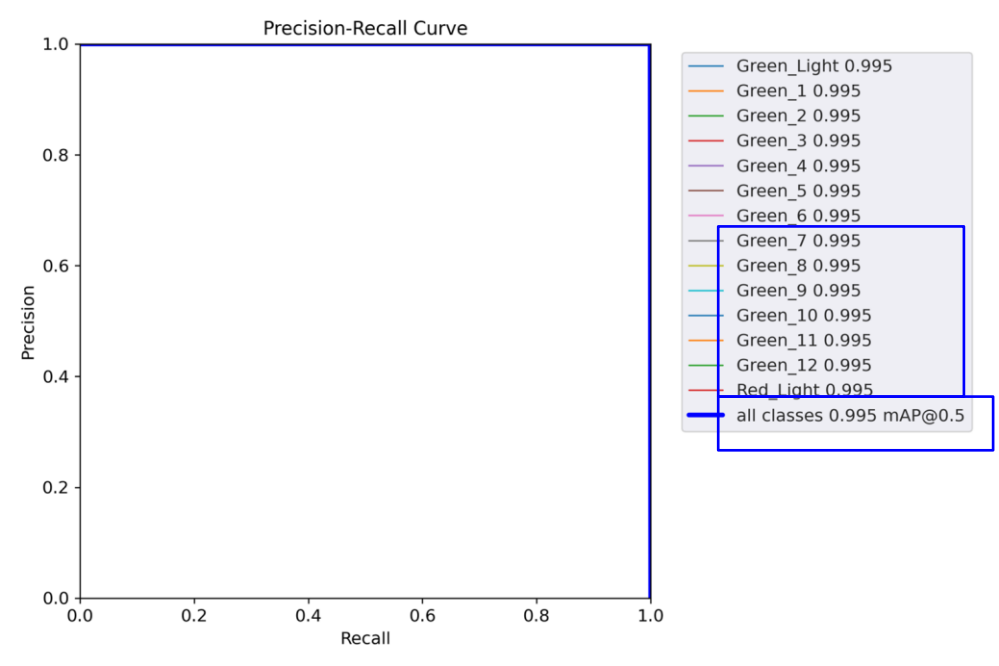
- **Noticed Data Imbalance**
- **Could lead to low accuracy of some classes**
  - **Because the class with higher # of data can be validated more than those that are not.**
  - **Then only that class could have more correct than other classes because other classes might be validated once and got wrong whereas that class might be validated twice and got wrong once and correct once.**
- **Therefore the # of data should be balanced.**

Experimental Results

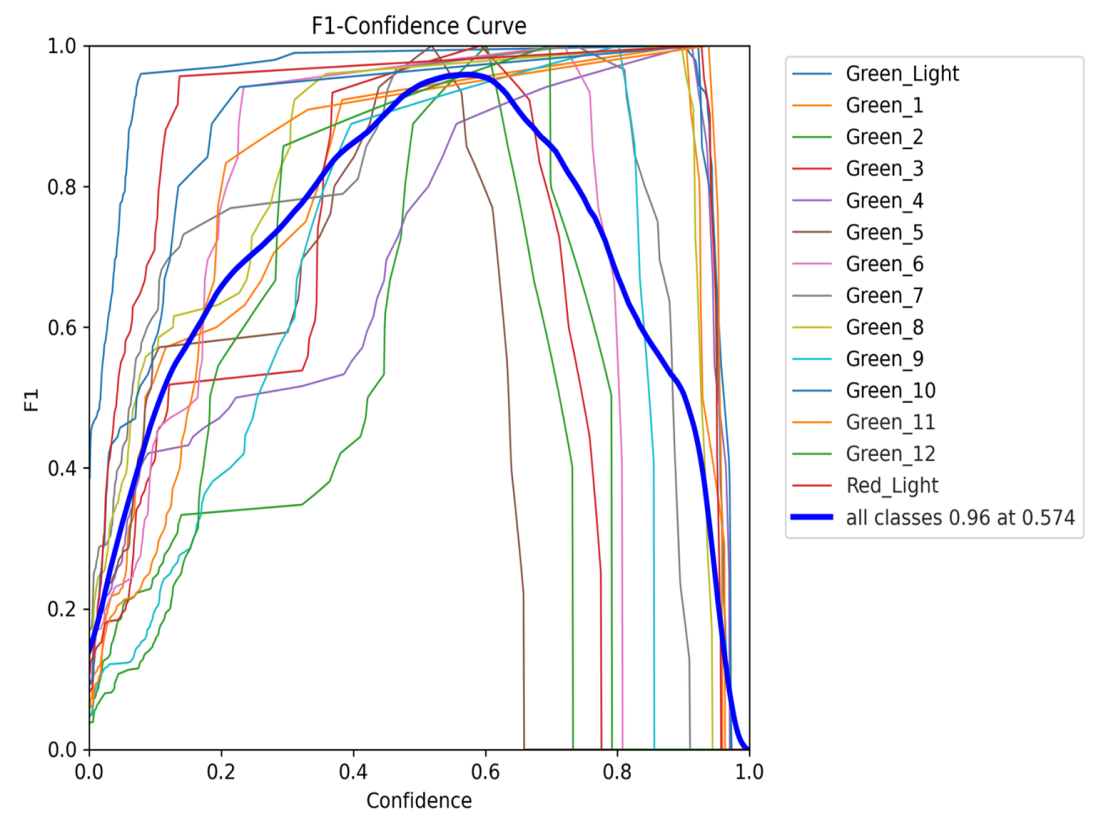
- Data Imbalance and Variation in Validation Size

- 7:3, 8:2, 9:1

The precision for every class at every point of recall is 1.0. The recall for every class at every point of precision is also 1.0. The left table shows that every class have been correctly matched by model by 99.5% chances. This shows the model improved and became accurate in recognizing the traffic light.



The F1(the precision + recall of the data) of all classes was 0.96(96%) when the confidence rate was 0.574(57.4%). Overall classes show a steady growth in F1 rate after it crosses about 0.5 confidence.



Experimental Results

- GreenEye Comparison of Data Imbalance and Data Balance

	Imbalanced Data	Balanced Data
Training Time (Hours)	2.39	**1.134**
mAP@0.5	**0.746**	**0.995**

- Comparison with Previous Research[4]

Traffic Colors

	Previous	GreenEye
Red Light	97.1	99.5
Green Light	100	99.5
Average	**98.52**	**99.5**

Traffic Number(1-12)

	Previous	GreenEye
Number	**None**	**99.5**

Experimental Results

- Traffic Light Recognition in Videos [Training:Validation = 9:1]

Conclusion

My goal was creating a real-time traffic light recognition model that can be used by the vision-disabled people when crossing the crosswalk. And I accomplished creating a **customized Yolo model on Google Colab that classifies 14 distinct traffic lights for both images and videos**.

Performance:

- **Initially with 884 data and 9:1 training to validation ratio, the average precision rate of my custom YOLO model was 0.746.**
  - **4 classes out of 14 had precision of <0.4. It turned out that Data Imbalance was causing this problem.**
- **Implementation to this problem was balancing the data to a range of approximately 70-80 images per class for most of the classes. By doing so, after balancing the data, my model that was trained with 9:1 of training to validation ratio had improved average precision rate of 0.995 and every class(14) had 0.995 precision rate.**

Future Applications

The model will be used in future via an app that vision-disabled people can use, which will be newly developed by me. When the vision-disabled user opens the app and start to use it, the app will allow the traffic light recognition in real-time by using the smartphone camera. If the user tries to walk across the crosswalk, the app will audibly read the current status of the traffic light - the color of the traffic light and the remaining seconds to walk across the crosswalk - using the TTS(text-to-speech) engine.